Technical Documentation

Product Line Profitability & Margin Performance Analysis for Nassau Candy Distributor

Detailed guide and project requirements for the Product Line Profitability & Margin Performance Analysis for Nassau Candy Distributor analysis.

Reference Websites

Unified Mentor: https://www.unifiedmentor.com/

Nassau Candy Distributor: https://www.nassaucandy.com/

This document preserves the complete content supplied in the project specification, with formatting and layout improvements only.

Background and Context

For distributors like Nassau Candy, sales volume alone is misleading. Some products:

- Sell in high volume but generate low profit
- Consume disproportionate manufacturing or operational cost
- Appear successful but weaken overall margins

To remain competitive, the company must understand:

- Which products truly drive profit
- Which divisions underperform financially
- Where pricing, sourcing, or product rationalization is needed

Problem Statement

Currently, the organization lacks visibility into:

- **Which product lines deliver the highest gross margin**
- **Whether high-sales products are actually profitable**
- **How profitability varies across product divisions**
- **Which products represent margin risk**

Without this insight, decisions on pricing, promotions, and product portfolio remain reactive and intuition-based.

Dataset Fields Description

FieldDescription	
Row ID	Unique row identifier
Order ID	Unique order identifier
Order Date	Date of order
Ship Date	Date of shipment
Ship Mode	Shipping method of order
Customer ID	Unique customer identifier
Country/Region	Country or region of customer
City	City of customer
State/Province	State/province of customer
Postal Code	Postal code / zip code
Division	Product division
Region	Region of customer
Product ID	Unique product identifier
Product Name	Product long namer
Sales	Total sales value of order
Units	Total units of order
Gross Profit	Gross profit of order (Sales - Cost)
Cost	Cost to manufacture

Analytical Methodology (Step-by-Step)

Data Cleaning & Validation

- Validate cost and sales values
- Remove zero-sales or invalid profit records
- Handle missing unit values
- Standardize product and division labels

Profitability Metric Calculation

For each product:

- Gross Margin (%)
- Profit per unit
- Total profit contribution

Product-Level Profitability Analysis

Rank products by:

- Gross profit
- Gross margin

Identify:

- High-profit / high-margin products
- High-sales / low-margin products
- Low-sales / low-profit products

Division-Level Performance Analysis

- Aggregate metrics by Division

Compare:

- Average margin by division
- Revenue vs profit imbalance

Identify divisions with:

- Strong financial efficiency
- Structural margin issues

Profit Concentration (Pareto) Analysis

Determine % of products contributing:

- 80% of revenue
- 80% of profit
- Detect congestion-prone states or regionIdentify over-dependency risks

Cost Structure Diagnostics

- Cost vs sales scatter analysis

Identify:

- Cost-heavy, margin-poor products
- Pricing inefficiencies

Flag products needing:

- Repricing
- Cost renegotiation
- Discontinuation review

Key Performance Indicators (KPIs)

KPIDescription	
Gross Margin (%)	Gross Profit ÷ Sales
Profit per Unit	Gross Profit ÷ Units
Revenue Contribution	Product sales ÷ total sales
Profit Contribution	Product profit ÷ total profit
Margin Volatility	Variability of margin over time

Streamlit Web Application Requirements

Dashboard Modules

• Product Profitability Overview

- Product-level margin leaderboard
- Profit contribution charts

• Division Performance Dashboard

- Revenue vs profit comparison
- Margin distribution by division

• Cost vs Margin Diagnostics

- Cost-sales scatter plots
- Margin risk flags

• Profit Concentration Analysis

- Pareto charts
- Dependency indicators

User Capabilities

- Date range selector
- Division filter
- Margin threshold slider
- Product search

Deliverables and Submission

- **Research paper (EDA, insights, recommendations)**
- **Streamlit dashboard (live analytics)**
- **Executive summary for government stakeholders**

Conclusion

This project establishes a clear, data-driven understanding of shipping route efficiency for Nassau Candy Distributor. By transforming raw order and shipment data into route-level operational intelligence, the organization gains actionable insights to improve logistics performance, reduce delays, and enhance nationwide delivery reliability.

Factories Co-ordinates

FactoryLatitudeLongitude		
Lot's O' Nuts	32.881893	-111.768036
Wicked Choccy's	32.076176	-81.088371
Sugar Shack	48.11914	-96.18115
Secret Factory	41.446333	-90.565487
The Other Factory	35.1175	-89.971107

Products and Factories Correlation

DivisionProduct NameFactory		
Chocolate	Wonka Bar - Nutty Crunch Surprise	Lot's O' Nuts
Chocolate	Wonka Bar - Fudge Mallows	Lot's O' Nuts
Chocolate	Wonka Bar -Scrumdiddlyumptious	Lot's O' Nuts
Chocolate	Wonka Bar - Milk Chocolate	Wicked Choccy's
Chocolate	Wonka Bar - Triple Dazzle Caramel	Wicked Choccy's
Sugar	Laffy Taffy	Sugar Shack
Sugar	SweeTARTS	Sugar Shack
Sugar	Nerds	Sugar Shack
Sugar	Fun Dip	Sugar Shack
Other	Fizzy Lifting Drinks	Sugar Shack
Sugar	Everlasting Gobstopper	Secret Factory
Sugar	Hair Toffee	The Other Factory
Other	Lickable Wallpaper	Secret Factory
Other	Wonka Gum	Secret Factory
Other	Kazookles	The Other Factory